Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 16)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Figure 1 shows the daily deaths in London and the mean concentration of smoke in the air during the smog of December 1952.

Figure 1

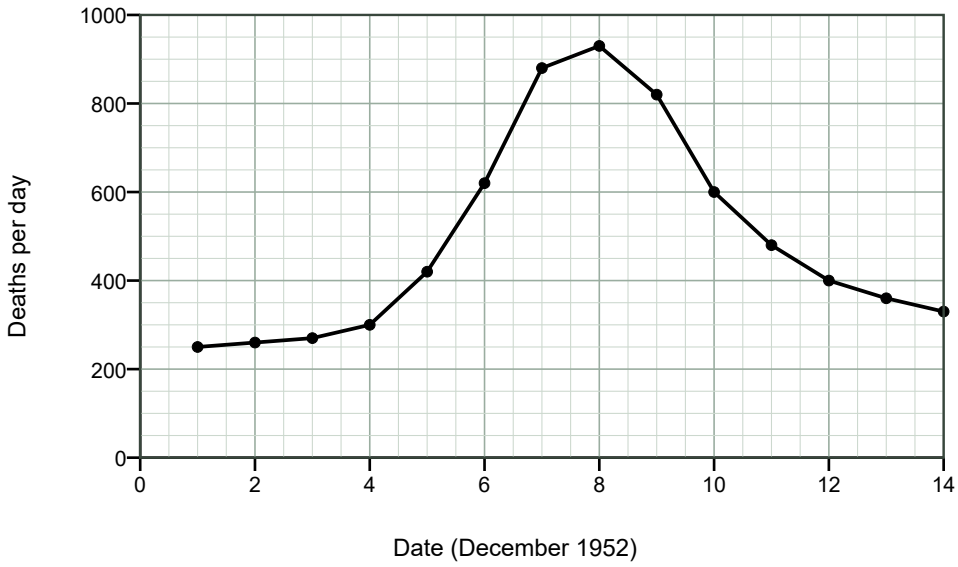


Table 1

Date	3 Dec	5 Dec	7 Dec	8 Dec	10 Dec	12 Dec
Smoke / mg m^{-3}	0.4	1.5	4.4	4.5	1.2	0.5

0 1 . 1 Describe the relationship between smoke concentration and daily deaths shown in Figure 1 and Table 1.

[2 marks]

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0 1 . 2 Calculate the percentage increase in daily deaths between 1 December and 8 December.

[2 marks]

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Answer ____ %

0 1 . 3 Explain how the weather conditions in December 1952 caused the smog to form.

[3 marks]

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0 1 . 4 Explain why most of the deaths were among the elderly, the very young and people with existing respiratory disease.

[2 marks]

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0 1 . 5 State the legislation introduced in the UK in response to this smog.

[1 mark]

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0 2 Table 1 shows the rate of heat loss through 1 m^2 of different glazing systems for a 1 °C temperature difference (U-value).

Table 1

Glazing	**Heat loss / W m^{-2} K^{-1}**
Single glazing	5.8
Double glazing, air filled	2.8
Double glazing, argon filled, low-e glass	1.6
Triple glazing, krypton filled, low-e glass	0.8
Double glazing, vacuum (0.2 mm gap)	0.74

0 2 . 1 A house has 20 m^2 of single glazing. The average temperature difference between inside and outside over the heating season is 12 °C. Calculate the reduction in the rate of heat loss if the windows are replaced with argon-filled low-e double glazing.

[3 marks]

Answer ..... W

0 2 . 2 Explain why the gap between the panes should be as large as possible but not too large.

[2 marks]

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0 2 . 3 Explain how low-emissivity glass and inert gas filling each reduce heat loss.

[2 marks]

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0 2 . 4 Explain why vacuum double glazing has a very narrow gap and where it is used.

[2 marks]

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0 2 . 5 State **one** reason why insulation thickness has an optimum value.

[1 mark]

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0 3 Table 1 shows global reactive nitrogen fixed each year by natural processes and by human activities.

Table 1

Source	Nitrogen fixed / million tonnes per year
Biological fixation (natural ecosystems)	110
Lightning	5
Haber process (fertiliser manufacture)	120
Cultivation of legume crops	60
Combustion (NOx)	30

0 3 . 1 Use Table 1 to calculate the percentage of total nitrogen fixation that is caused by human activities.

[2 marks]

Answer ..... %

0 3 . 2 Describe the Haber process and explain why it is unsustainable compared with biological nitrogen fixation.

[3 marks]

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0 3 . 3 Describe **three** consequences of the increased movement of nitrogen compounds caused by human activities.

[3 marks]

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0 3 . 4 Explain how composting or anaerobic digestion of biological wastes produces a better fertiliser than raw waste.

[2 marks]

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0 4 A factory discharges a persistent, water-soluble pollutant into a river. Table 1 shows the concentration of the pollutant measured at sites along the river on a day with normal river flow.

Table 1

Site	Distance downstream of discharge / km	Pollutant concentration / mg dm^{-3}	River flow / $m^3 s^{-1}$
A (upstream)	-2	0.0	12
B	0.5	4.8	12
C	5	3.9	15
D (below tributary)	12	1.6	36
E	30	1.4	38

0 4 . 1 Calculate the percentage decrease in pollutant concentration between site B and site E.

[2 marks]

Answer ☐ %

0 4 . 2 Explain the large decrease in concentration between sites C and D.

[2 marks]

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0 4 . 3 Explain why the concentration changes very little between sites D and E.

[2 marks]

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0 4 . 4 Suggest why the pollution might be more serious if the same discharge entered a lake instead of a river.

[2 marks]

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0 4 . 5 Explain why the factory might be permitted to discharge the effluent into the sea near a strong current but not into the river.

[2 marks]

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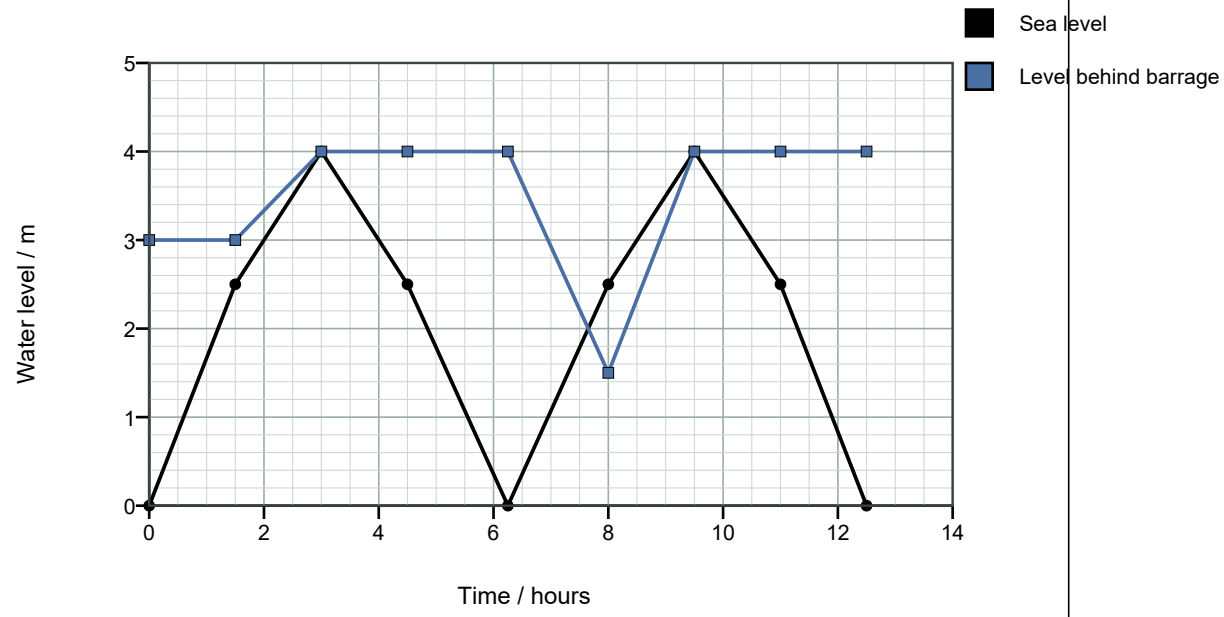
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0 5 Figure 1 shows the water level inside and outside a tidal barrage over one 12.5-hour tidal cycle and the periods when electricity is generated.

Figure 1



0 5 . 1 Use Figure 1 to explain when electricity is generated by the barrage.

[2 marks]

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0 5 . 2 Explain why tidal power is intermittent but predictable.

[2 marks]

0 5 . 3 Explain the difference between spring and neap tides and why this affects output.

[2 marks]

0 5 . 4 Explain how a tidal lagoon can be used to generate electricity when there is no tidal flow.

[2 marks]

0 5 . 5 Give **two** coastal features that make a site suitable for tidal power.

[2 marks]

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0 6 . 1 Explain how hydrogen can be used to store energy from intermittent renewable resources, and describe **two** ways the stored energy can be released.

[4 marks]

0 6 . 2 Compare **three** methods of storing hydrogen, giving a disadvantage of each.

[3 marks]

0 6 . 3 Explain how a fuel cell differs from a rechargeable battery.

[2 marks]

0 6 . 4 Explain what is meant by the 'hydrogen economy'.

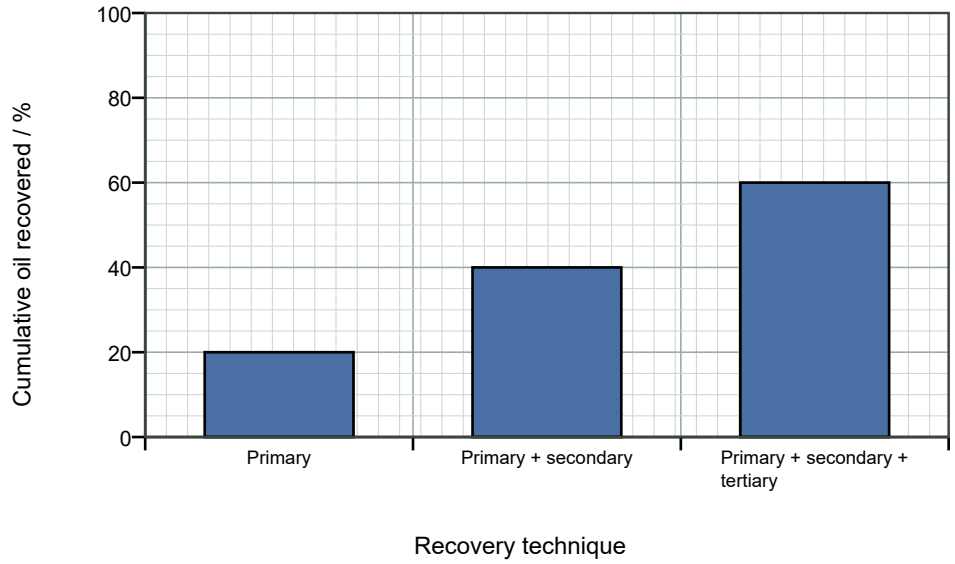
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0 7 Figure 1 shows the percentage of the oil in a reservoir that can be recovered using different recovery techniques.

Figure 1



0 7 . 1 Describe how primary oil recovery works and explain why it recovers only about 20% of the oil.

[2 marks]

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0 7 . 2 Describe secondary recovery and explain how it can be linked to carbon capture and storage.

[2 marks]

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0 7 . 3 Describe **three** tertiary (enhanced) recovery techniques.

[3 marks]

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0 7 . 4 Explain **three** advantages of directional drilling.

[3 marks]

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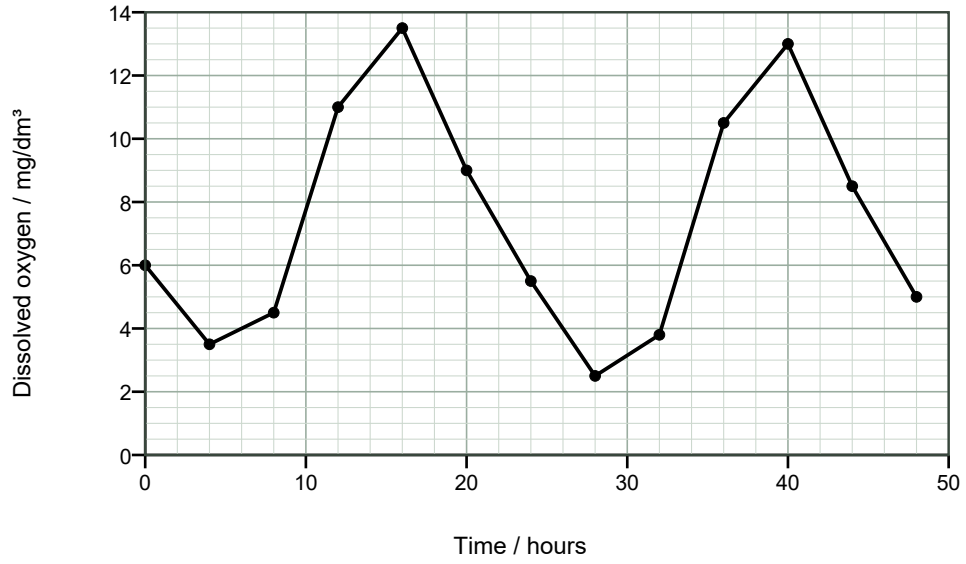
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0 8 Figure 1 shows the dissolved oxygen concentration in a shallow lowland lake over 48 hours in late summer. The lake receives runoff from arable farmland and effluent from a sewage works.

Figure 1



0 8 . 1 Describe the pattern shown in Figure 1.

[2 marks]

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0 8 . 2 Explain the causes of the pattern shown in Figure 1.

[4 marks]

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0 8 . 3 Explain why fish kills in eutrophic lakes usually happen at night or at the end of summer.

[2 marks]

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0 8 . 4 Give **two** other harmful effects of algal blooms.

[2 marks]

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0 9 . 1 A student measured the pH of a stream with a meter that reads to 0.01 pH units but had not been calibrated. Which statement is correct? Tick **one** box.

[1 mark]

A The readings are precise but may not be accurate ☐

B The readings are accurate but not precise ☐

C The readings are both accurate and precise ☐

D The readings are reliable because the meter is electronic ☐

0 9 . 2 Explain what is meant by an **anomalous result** and how a student should decide whether to include it.

[2 marks]

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0 9 . 3 Explain why a preliminary study is carried out before the main investigation.

[2 marks]

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1 0 . 1 Ozone depletion is often described as an environmental problem that has been tackled successfully. Compare the effectiveness of the methods used to restore the ozone layer with the effectiveness of attempts to tackle global climate change, and explain the reasons for the difference.

[9 marks]

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Extra space

1 0 . 2 Name the international agreement of 1985 that preceded the Montreal Protocol and state its main weakness.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.

[25 marks]

OR

1 1 . 2 Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS